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Wind load Analysis For Rolling Shutter

as per Indian Standards IS 875 Part III 1987

Project: ALSTOM
Client: TATA PROJECT
Date: 08/06/2017
Height: 6000 mm
Width: 4000 mm
Slat Size: 95 mm
Slat Width: 19 mm

Wind Load Analysis as per IS

Design Wind Speed Vz= Vb * k1 * k2 * k3

where Vb = Basic Wind Speed in m / s.	=	47 m/s
k1 = Risk Coefficient (clause 5.)	=	1
k2 = THS Factor (clause 5.3	=	1
k3 = Topography Factor (clause 5.)	=	1
Vz = Design Wind Speed	=	47 m/s

Terrain Category: 2

(Open terrain with well-scattered obstructions having heights between 1.5m to 10m)

Class of Building: A(Structures having max. dimensions is less then 20m)

k1, k2 & k3 constants are derived from IS-875, based on location height of structure & geographical factor. These factors are required for tall structures, Where we need to consider varying pressure condition.

Wind Load Analysis as per IS 875 - III

Design wind pressure = $0.6 \times V_z^2$

Where,

Pz = Design wind pressure

Vz = Design wind velocity

Wind Pressure, Pz = 0.6 x (47)2

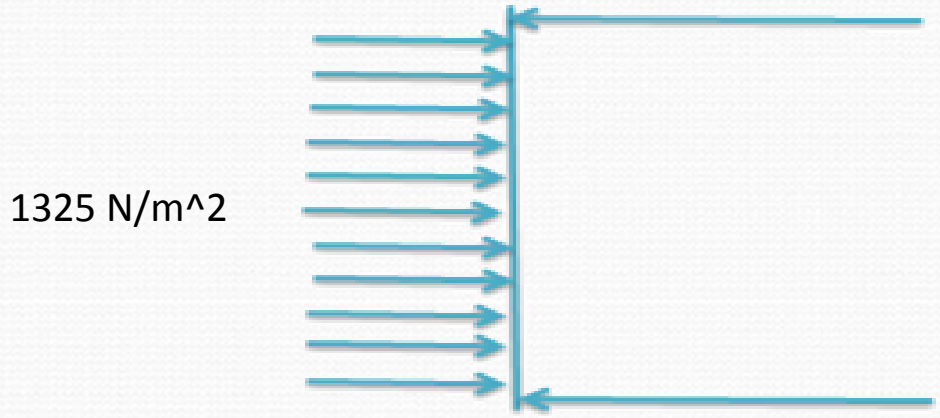
Wind Pressure, Pz = 1325.4 N/m^2

Pz = 0.001325 N/mm^2

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Wind Load Analysis as per IS 875 - III

Assumed - Simply supported beam with uniform pressure.



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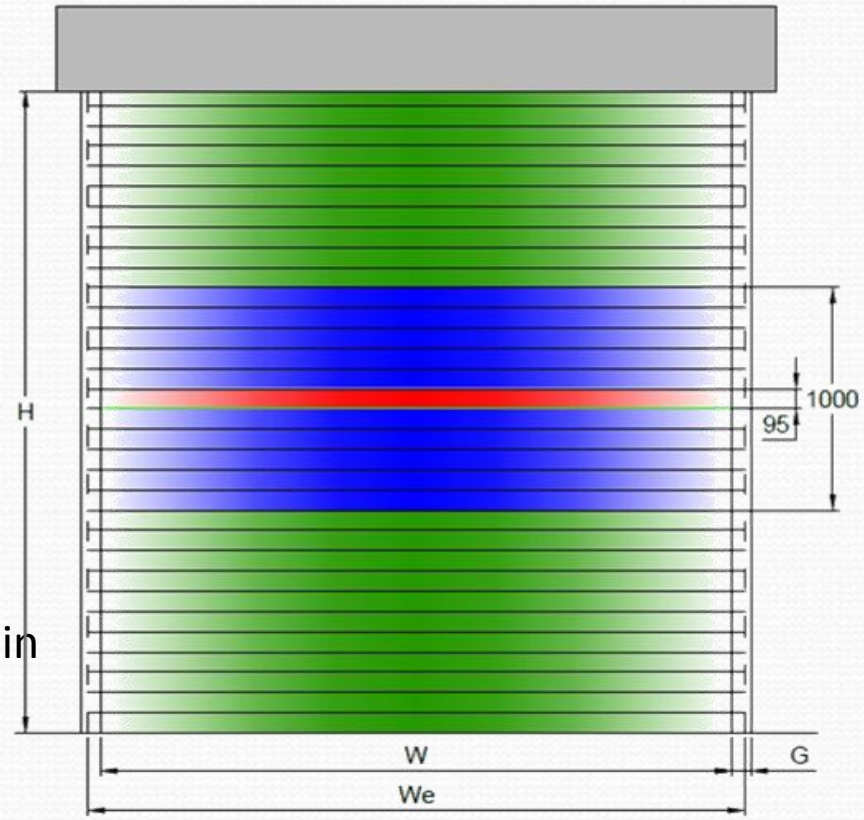
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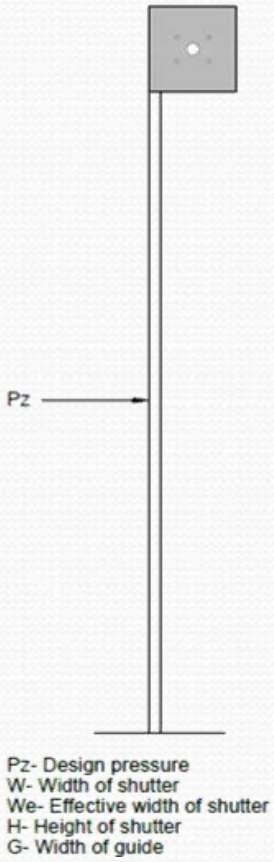
A)Bending Moment

B)Deflection

C)Maximum Stress

D)Side guides for curtain





Pz- Design pressure
W- Width of shutter
We- Effective width of shutter
H- Height of shutter
G- Width of guide

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A)Bending Moment

Door is simply supported on two opposite sides

Max B.M. $M = \alpha \times W \times L^2/8$

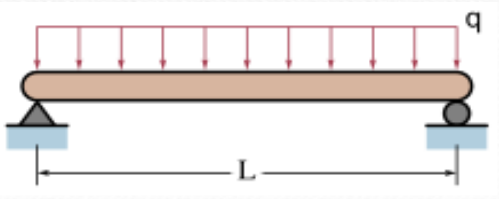
Where, W= UDL load

α= Bending moment coefficient as per IS 456:2000-Table26

B)Deflection

Max. deflection will be at the centre of slat

D max = (5x M x L^2) / (48 x E x I)



Where ,

M = Bending Moment of Slat

E = Modules of Elasticity for CRS = 210GPa (KN/mm2)

I = Moment of Inertia of slat (From CAD Data)

L = Clear width of Shutter

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C)Max Stress

Max. Stress= M x $\bar{y}$/ I

Where,

M=Bending moment of slat, $\bar{y}$= dist of side on y-y from CG , I= moment of inertia ,

Permissible Stress =248/FOS

=248/1.5

= 165 N/sq.mm

D)Side Guides for Curtain panel

Side Guides required = w/60

Where,

w= width of shutter

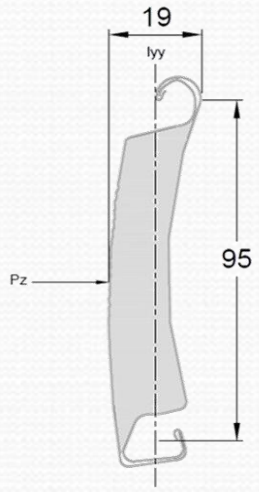
Design calculations for shutter of
Size: 4000mm(w) x 6000mm(H)

1. Wind Analysis for rolling shutter of size 4000mm(w) x 6000mm(h)

Design Wind Speed Vz= Vb * k1 * k2 * k3

where,

Vb = Basic Wind Speed in m / s.	=	47
k1 = Risk Coefficient (clause 5.4.1)	=	1
k2 = THS Factor (clause 5.3.2)	=	1
k3 = Topography Factor (clause 5.3.3)	=	1
Vz = Design wind Speed	=	47 m/s



Design wind pressure = 0.6x 47x47= 1325.4 N/m2

UDL acting on Slat, W = Pz x A/L Where, A= Area of slat

= 0.0013254 x (4000 x 95) / 4000

W = 0.1259 N/mm

Bending Moment $M = \alpha.(W \times L^2)/8$

= 0.089 x 0.1259 x 4000 x 4000 / 8

= 22410.20 N-mm

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Ø Assuming thickness of slat 0.5 mm & minimum height as 95mm & depth as 19mm.

(From CAD DATA) I= 8450 mm^4

E= modulus of elasticity for CRS= 210 GPA

= 210000 N/mm^2

Max. deflection = 5 x 22410.2 x 4000 x 4000 / (48 x 210000 x 8450)

= 21.05 mm

Permissible deflection = 4000/150

As per IS:800 = 26.667 mm

Deflection is within limit.

Max. stress = 22410.2x19/(2x8450)

= 25.19 N/mm^2

Permissible stress = (248 N/mm2) /1.5

= 165 N/mm^2

Note - As max. stress is less than permissible stress, hence no permanent deformation of slat will occur.

Guide side required = 4000/60 = 66.67

Hence we are using 100mm guide.

Thank You

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